Anis-Lounes TAZEROUT

Control Engineer — M2 Student in Advanced Systems and Robotics

• anis.tazerout@hotmail.com • +33 (0)7 58 99 55 63 • LinkedIn: Anis-Lounes Tazerout • GitHub: AnisTaz

Profile

Research-oriented engineer with a solid background in control theory, applied mathematics, signal processing, and numerical optimisation. My current research focuses on the benchmarking, comparison, and improvement of multi-robot collaborative SLAM (C-SLAM) frameworks, with state estimation, sensor fusion, and pose graph optimisation at the centre of my daily work. Co-author of an experimental study on quantum-inspired signal processing for embedded control. Strongly motivated to pursue a PhD at the intersection of estimation and control applied to robotics, with a particular interest in aerial robotics, observer design, and the modelling and control of interactive autonomous systems.

Education

M2 — Advanced Systems and Robotics (SAR) *2025 – ongoing*
Sorbonne Université, Paris, France

State Engineering Degree (MSc equivalent) — Control Engineering *2022 – 2025*
École Nationale Polytechnique d'Alger (ENP), Algeria *Ranked 19/1723 at the National Entrance Examination*

Preparatory Classes for Engineering Schools — Mathematics, Physics, and Chemistry *2020 – 2022*
École Nationale Polytechnique d'Alger (ENP), Algeria

Research Experience

Research Intern — *Benchmarking, comparison, and improvement of multi-robot collaborative SLAM frameworks* *March 2026 – August 2026*
ISPR, Institut Pascal, Clermont-Ferrand, France — Supervisors: Dr. Paul Checchin and Dr. Ahmad K. Aijazi

- Systematic state of the art of multi-robot collaborative SLAM (C-SLAM) algorithms, covering both front-end (LiDAR odometry, place recognition, descriptor-based loop closure) and back-end (distributed pose graph optimisation, outlier rejection) stages.
- Comparative analysis of LiDAR-based C-SLAM frameworks (Swarm-SLAM, DCL-SLAM, maplab 2.0), including multi-modal variants integrating GNSS and IMU, evaluated on public datasets using the `evo` toolkit for trajectory and accuracy metrics.
- Design of reproducible benchmarking pipelines under ROS 2 and Docker, with attention to multi-sensor data handling and real-time processing constraints.
- Formulation of concrete contribution directions aimed at improving the front-end or back-end of an open-source C-SLAM framework.

Research Project — *Quantum-inspired signal processing for embedded control* *February 2025 – June 2025*
Process Control Laboratory (LCP), École Nationale Polytechnique d'Alger — Supervisors: Prof. M. Tadjine (ENP) and Prof. N. Zioui (UQTR, Canada)

- Co-design of a quantum-inspired pulse-width modulation algorithm (QPWM) based on a novel real-valued quantum comparator, bridging theoretical formulation and numerical implementation.
- Implementation on an embedded platform driving a DC motor test bench in closed loop, and full experimental validation against the classical scheme, demonstrating measurable gains in precision and reference tracking.
- Co-author of a peer-reviewed scientific publication (see *Publications*).

Academic Projects

Modelling and robust control of a quadrotor
Derivation of kinematic and dynamic models of a 6-DoF underactuated aerial vehicle in MATLAB/Simulink.

Synthesis and comparative robustness study of advanced control laws (fuzzy logic, sliding mode, Lyapunov-based design), giving direct experience with the dynamics and control of UAV-type platforms.

MPC simulation of a manipulator arm on a mobile cart
Synthesis and simulation-based evaluation in Python (`do-mpc` library) of a Model Predictive Control law for the optimal trajectory control of a mobile manipulator under constraints, providing hands-on experience with optimisation-based control, horizon formulation, and real-time constraint handling.

URDF modelling and simulation-based control of a manipulator arm
Creation of the kinematic model (URDF) and implementation of the control chain in the ROS 2 and Gazebo simulation environment.

Kinematic modelling and control of a differential-drive robot
Development of several advanced control methods (fuzzy logic, gain scheduling, sliding mode) for trajectory tracking and line following, validated by simulation, providing hands-on experience with mobile-robot navigation.

Teaching

Teaching Assistant (*Travaux Dirigés*) *2024 – 2025*
Numidia Institute of Technology — Tutorial classes in Digital Systems for first-year (L1) undergraduate students.

Publications

Saidat, S., Boumekhita, R., **Tazerout, A.-L.**, Kouici, A., Fazilat, M., Tadjine, M., & Zioui, N. (2025). Experimental assessment of quantum PWM performance in DC motor speed control applications. *Proceedings of the 2025 International Conference on Decision Aid Sciences and Applications (DASA).* DOI:10.1109/DASA68193.2025.11498904

Technical Skills

Programming: Python (NumPy, SciPy, pandas, Matplotlib, `evo`, `do-mpc`, PyTorch), C++, MATLAB/Simulink.
Estimation and sensor fusion: Kalman filtering (KF, EKF, UKF), particle filtering, Bayesian estimation, MAP estimation, pose graph optimisation, factor graphs, probability and statistics.
Systems and control: dynamical systems analysis, observer synthesis, optimal control (MPC), robust control (H-∞), nonlinear control (sliding mode, Lyapunov-based design).
Numerical optimisation: convex and nonlinear optimisation, MPC formulation, Gauss-Newton and Levenberg-Marquardt for nonlinear least squares.
Robotics and perception: ROS 1 and ROS 2, Gazebo, kinematic and dynamic modelling (URDF), multi-sensor fusion (LiDAR, camera, IMU, GNSS), single- and multi-robot SLAM, classical path planning (A*, RRT*, artificial potential field).
Machine learning: working knowledge of PyTorch and scikit-learn; basics of reinforcement learning and evolutionary algorithms (DEAP).
Environments and tooling: Linux, Docker, Git/GitHub, LaTeX/BibTeX, Bash.

Languages

• English — C1 (EF SET certified) • French — fluent (daily, professional and academic use) • Arabic — native / bilingual